ITO / PELTIER CONTROLLER

24 V | XIAO nRF52840 | 4 TEC channels | ITO heater | 4 A prototype

POWER AND PROTECTION

02 R7B / input / eFuse

File: sheet_02.kicad_sch

03 Relay / emergency stop

File: sheet_03.kicad_sch

04 Heater / current sensing

File: sheet_04.kicad_sch

CONTROL AND CURRENT SETPOINTS

01 MCU / 5V / USB

File: sheet_01.kicad_sch

05 DAC / op-amp buffers

File: sheet_05.kicad_sch

06 TEC enables

File: sheet_06.kicad_sch

FOUR MATCHING TEC CURRENT REGULATORS

07 TEC 1

File: sheet_07.kicad_sch

08 TEC 2

File: sheet_08.kicad_sch

09 TEC 3

File: sheet_09.kicad_sch

10 TEC 4

File: sheet_10.kicad_sch

TEMPERATURE, FAN AND AMBIENT

11 Glass NTCs / ADC

File: sheet_11.kicad_sch

12 Heatsink / fan / I2C

File: sheet_12.kicad_sch

13 Remote SHT31 ports

File: sheet_13.kicad_sch

Pages 02–14: wired circuits within each block. Matching power symbols / sheet ports are connected.

ERC source declarations for the power rails:

PWR_FLAG GND

PWR_FLAG +24_IN

PWR_FLAG +24_PROT

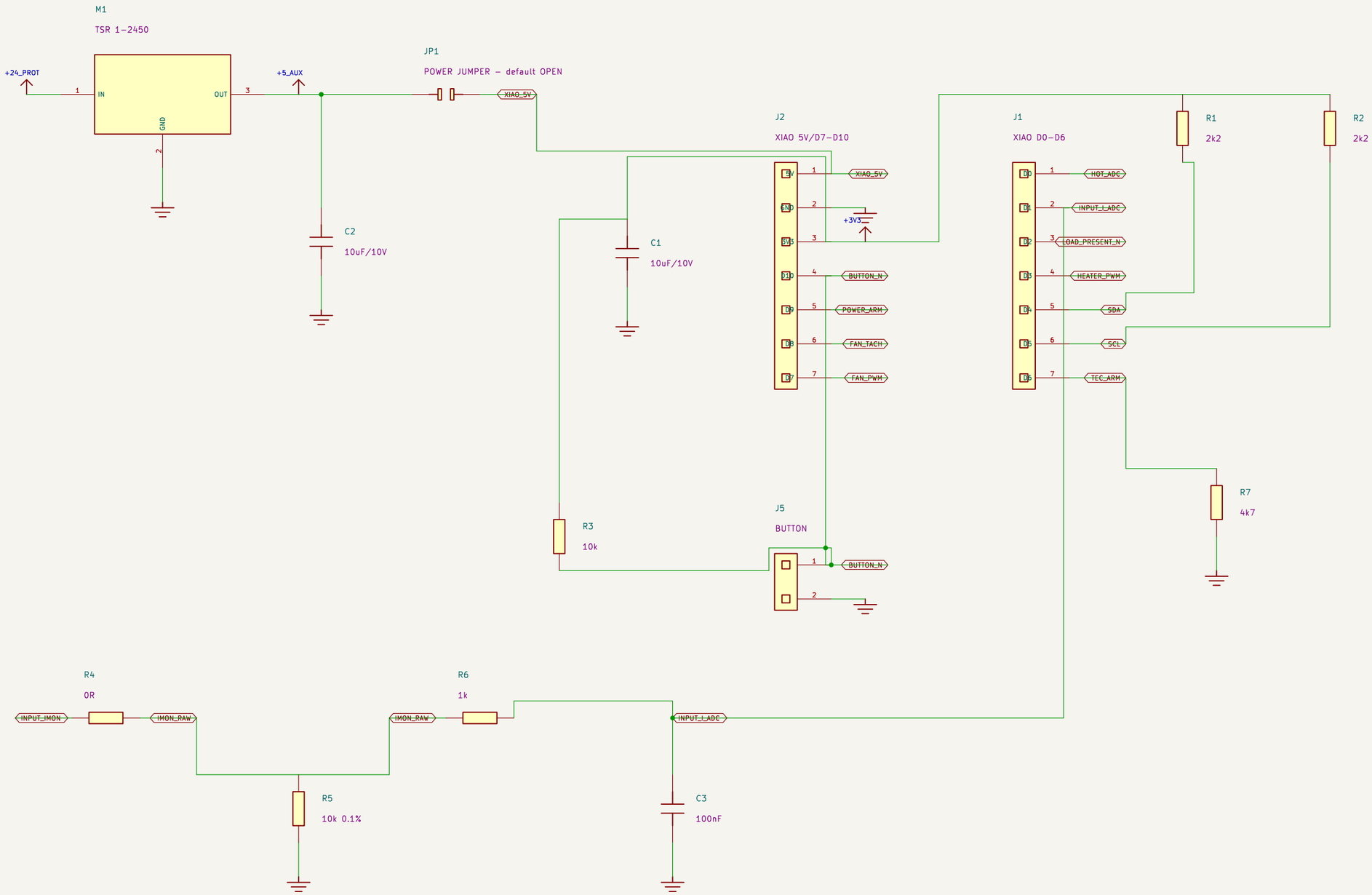
PWR_FLAG +24_LOAD

PWR_FLAG +5_AUX

PWR_FLAG +3V3

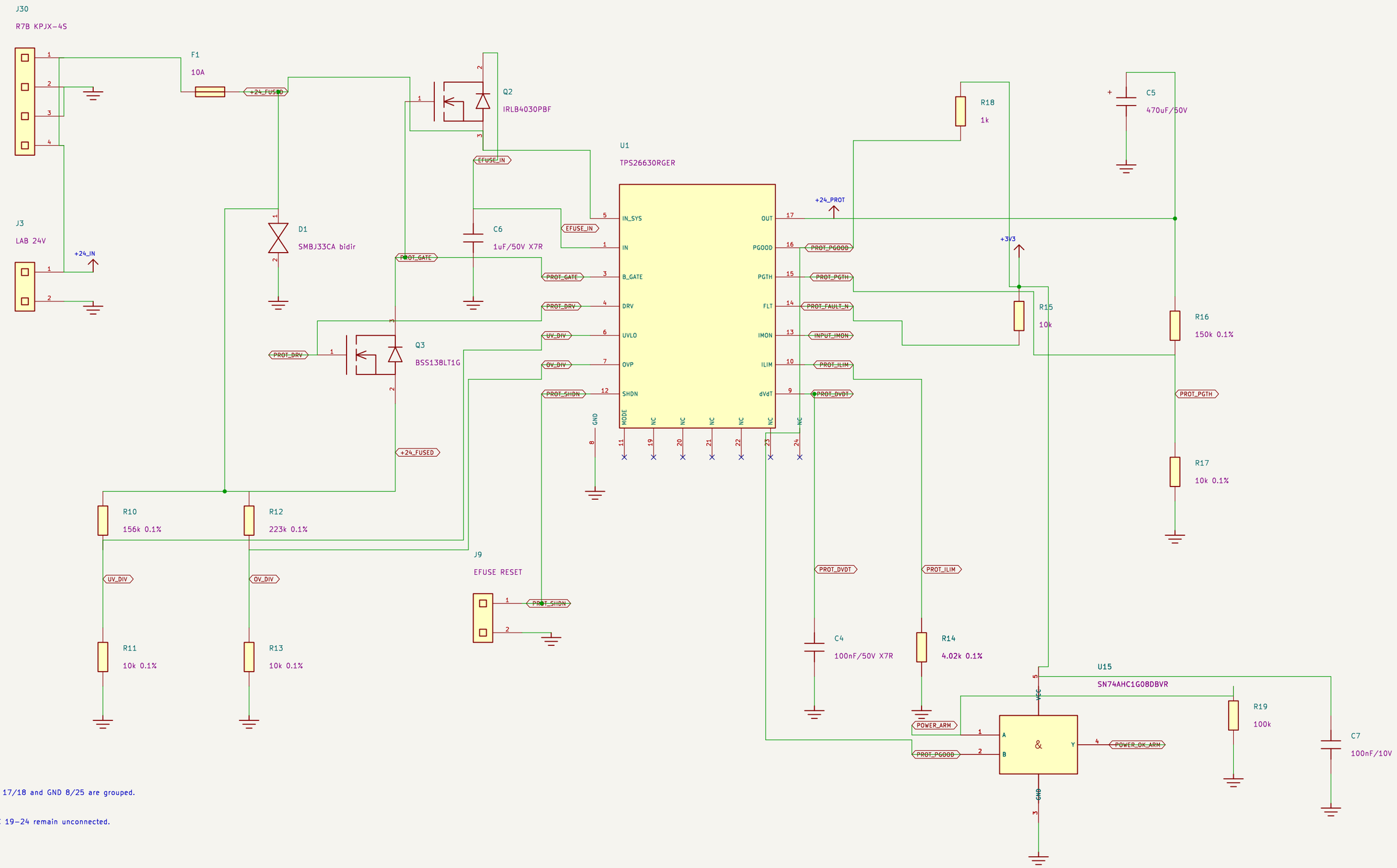
01 MCU and logic power

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB



02 Input protection

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB

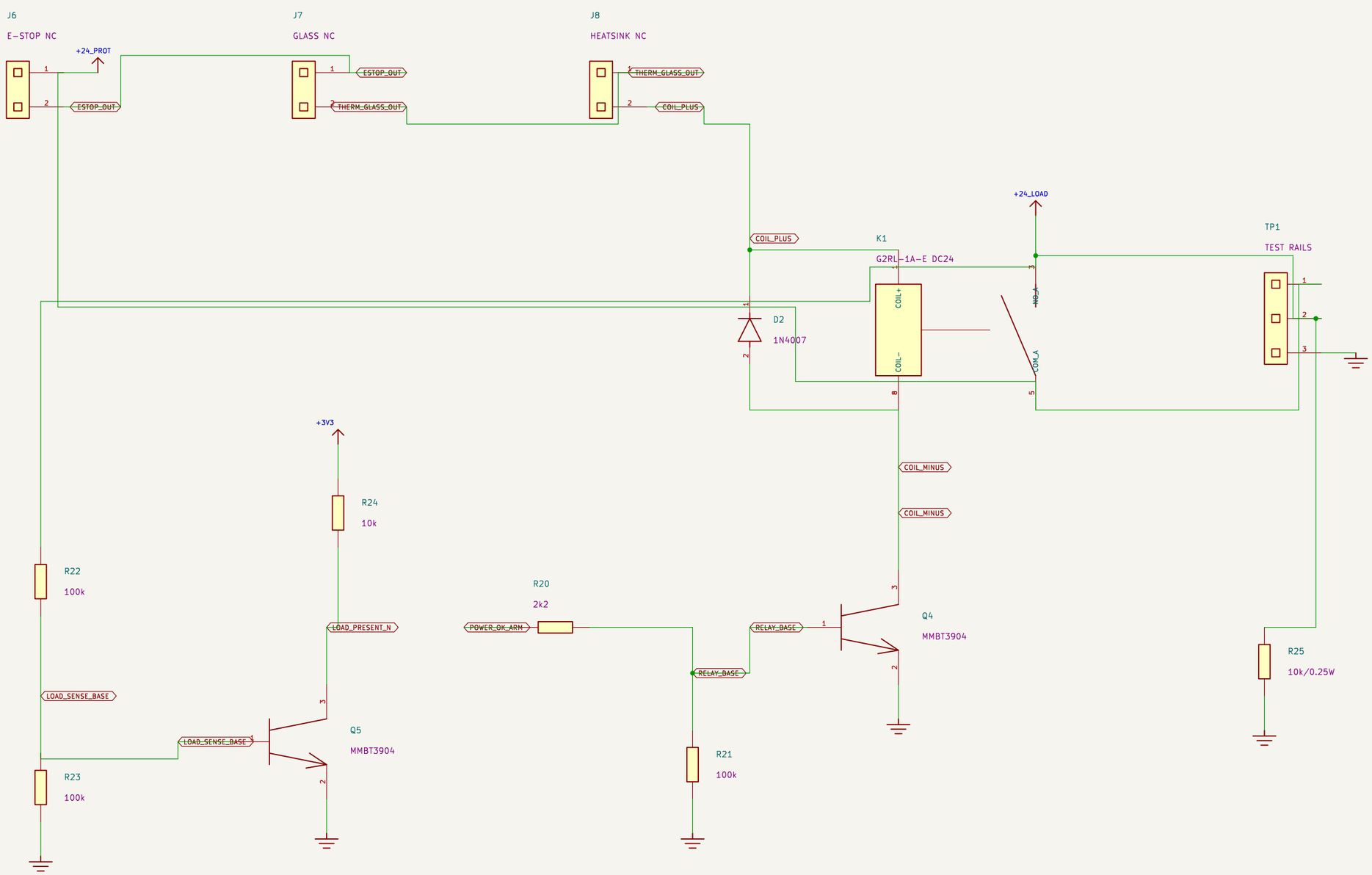


U1: IN 1/2, OUT 17/18 and GND 8/25 are grouped.

MODE 11 and NC 19–24 remain unconnected.

03 Relay and hardware shutdown

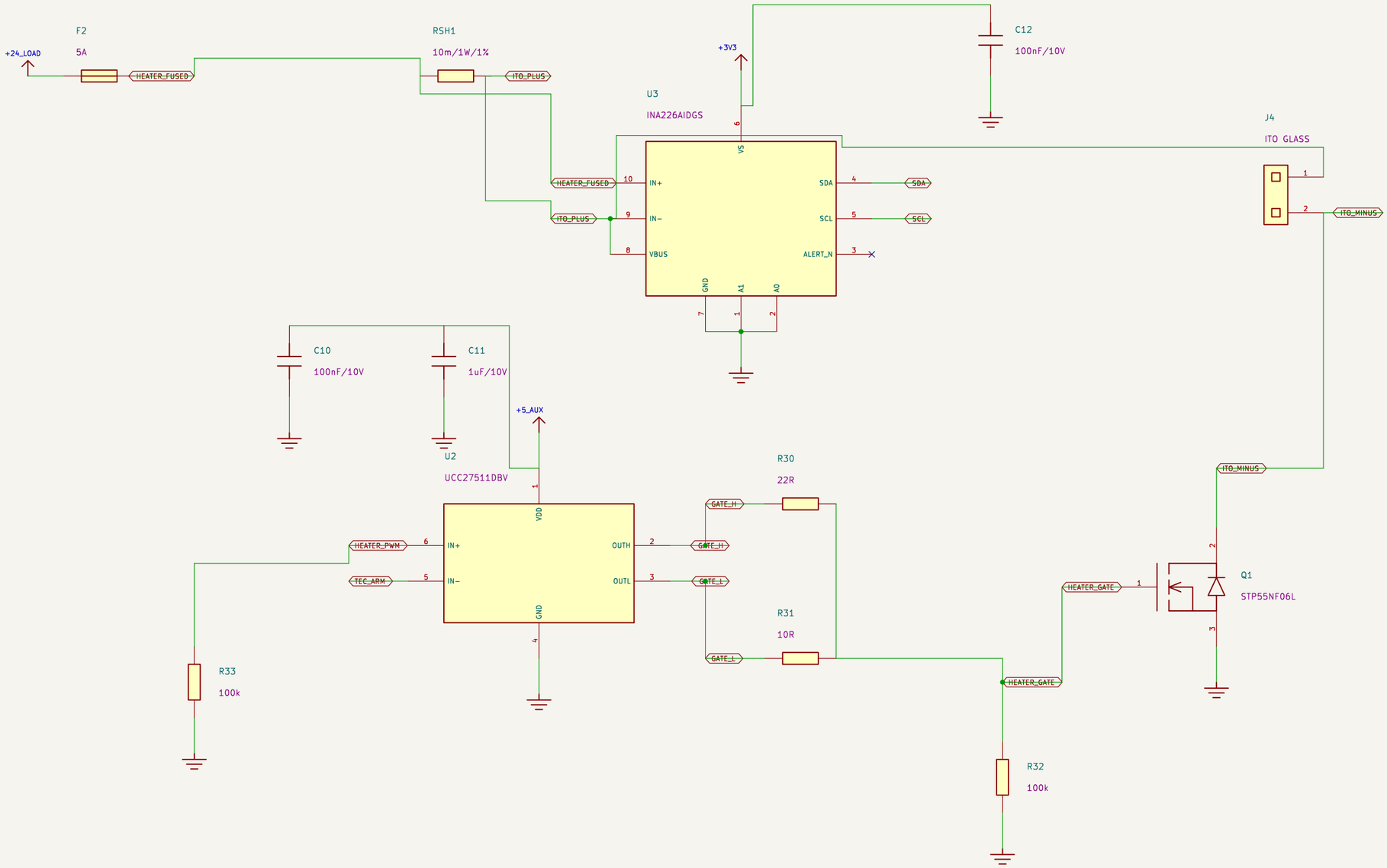
ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB



K1: COM 5/6 and NO 3/4 in parallel; coil on 1/8.

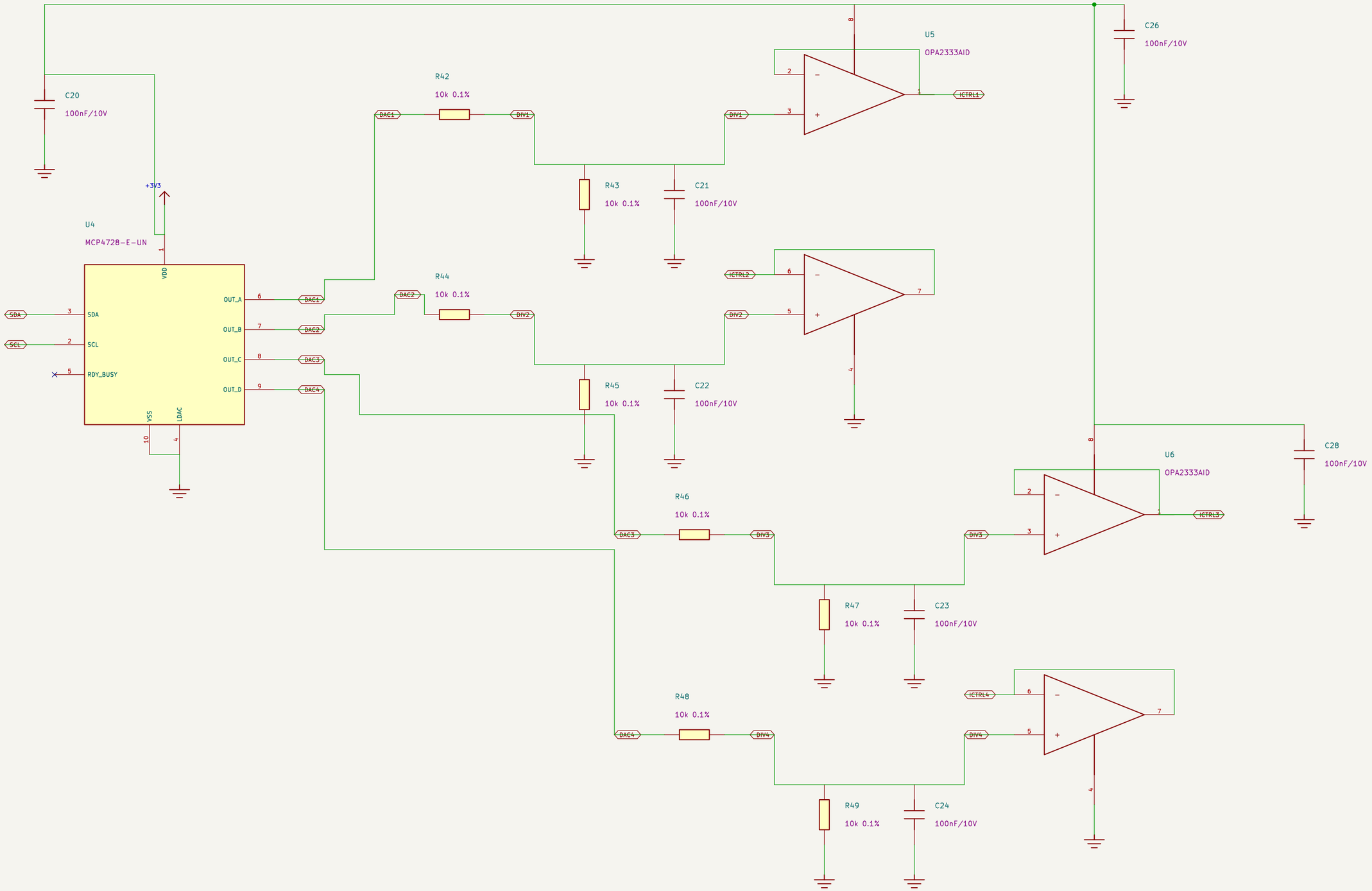
04 ITO heater

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB



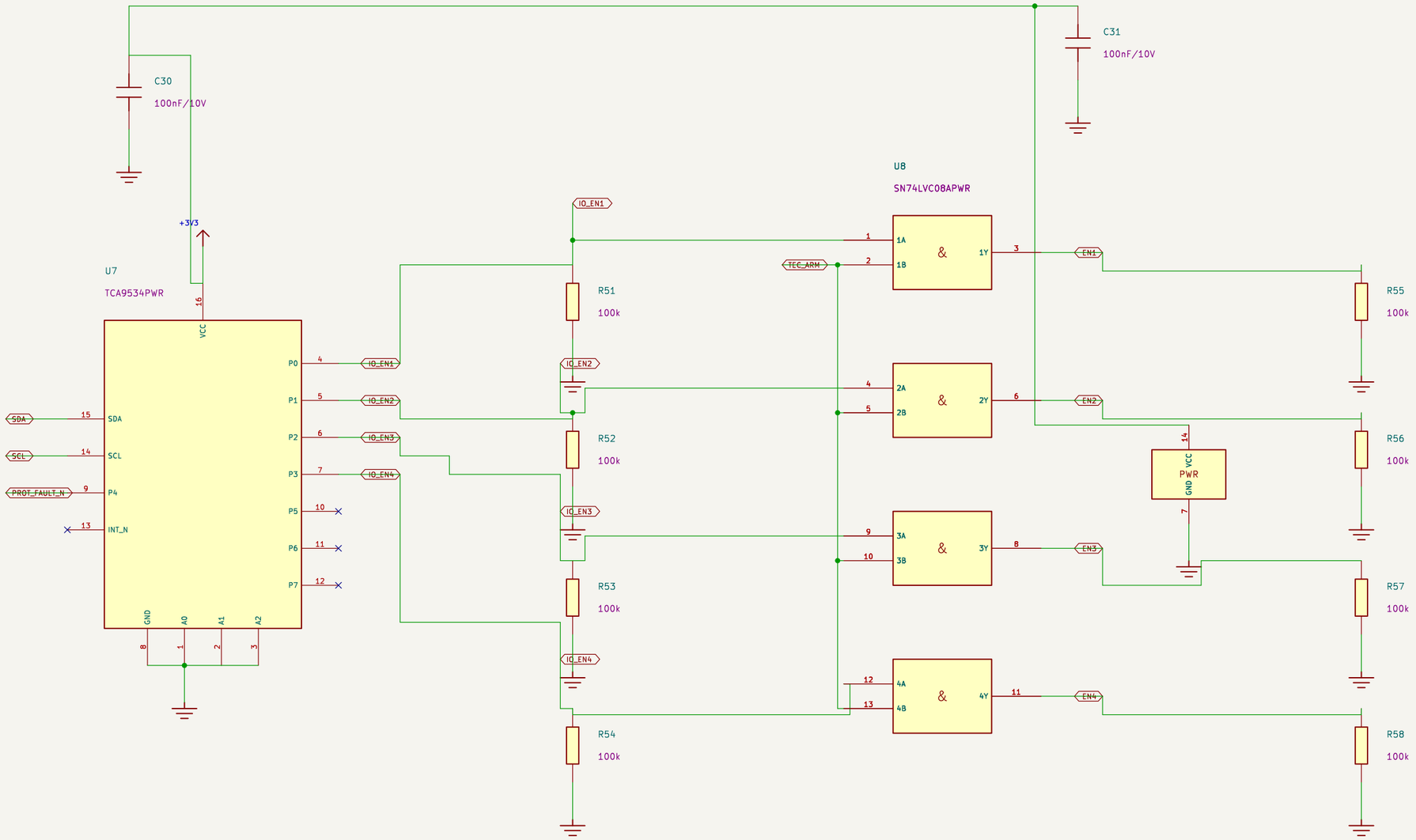
05 DAC and current setpoints

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB



06 TEC channel enables

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB

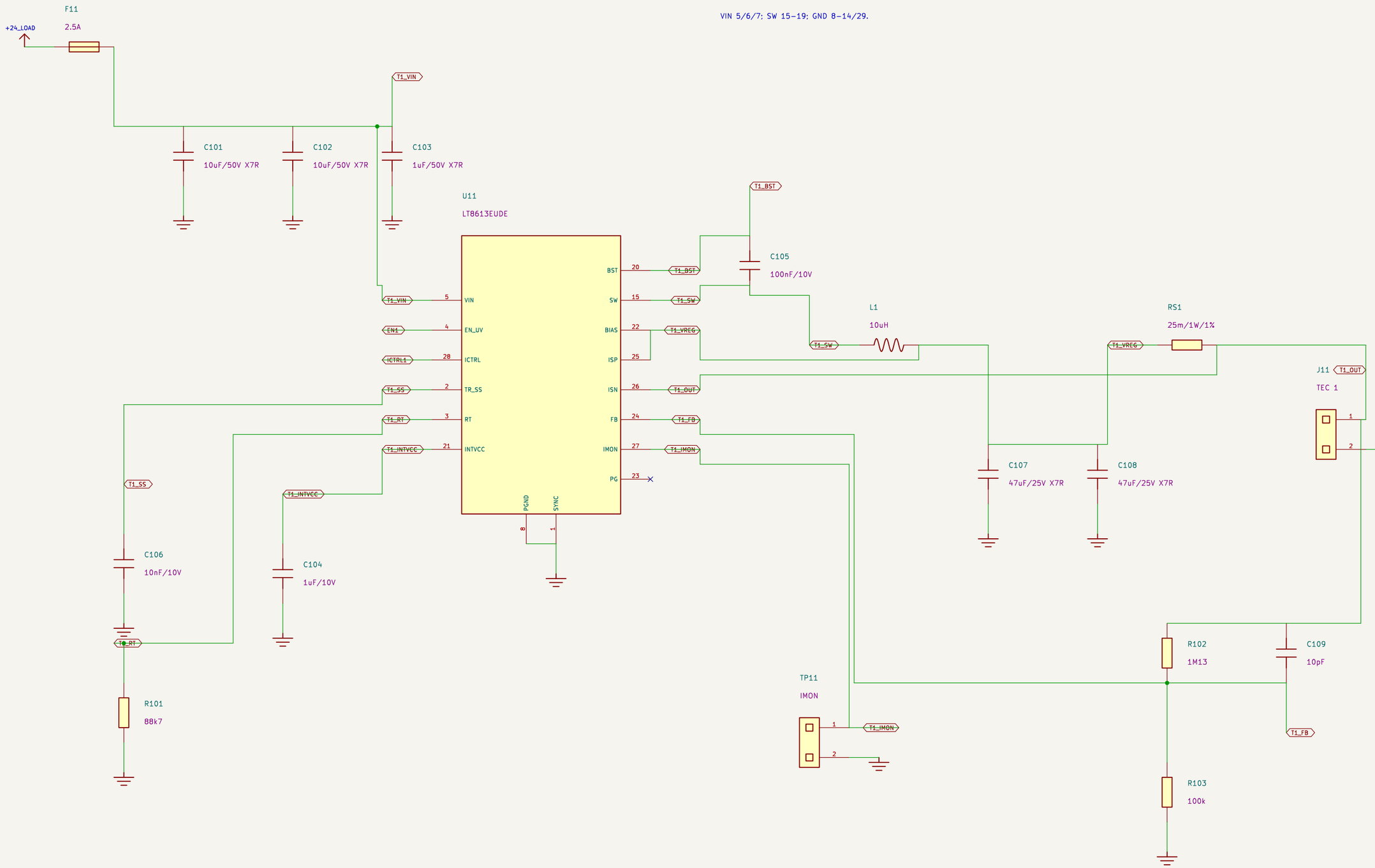


07 TEC current regulator 1

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB

Grouped IC pins:

VIN 5/6/7; SW 15–19; GND 8–14/29.

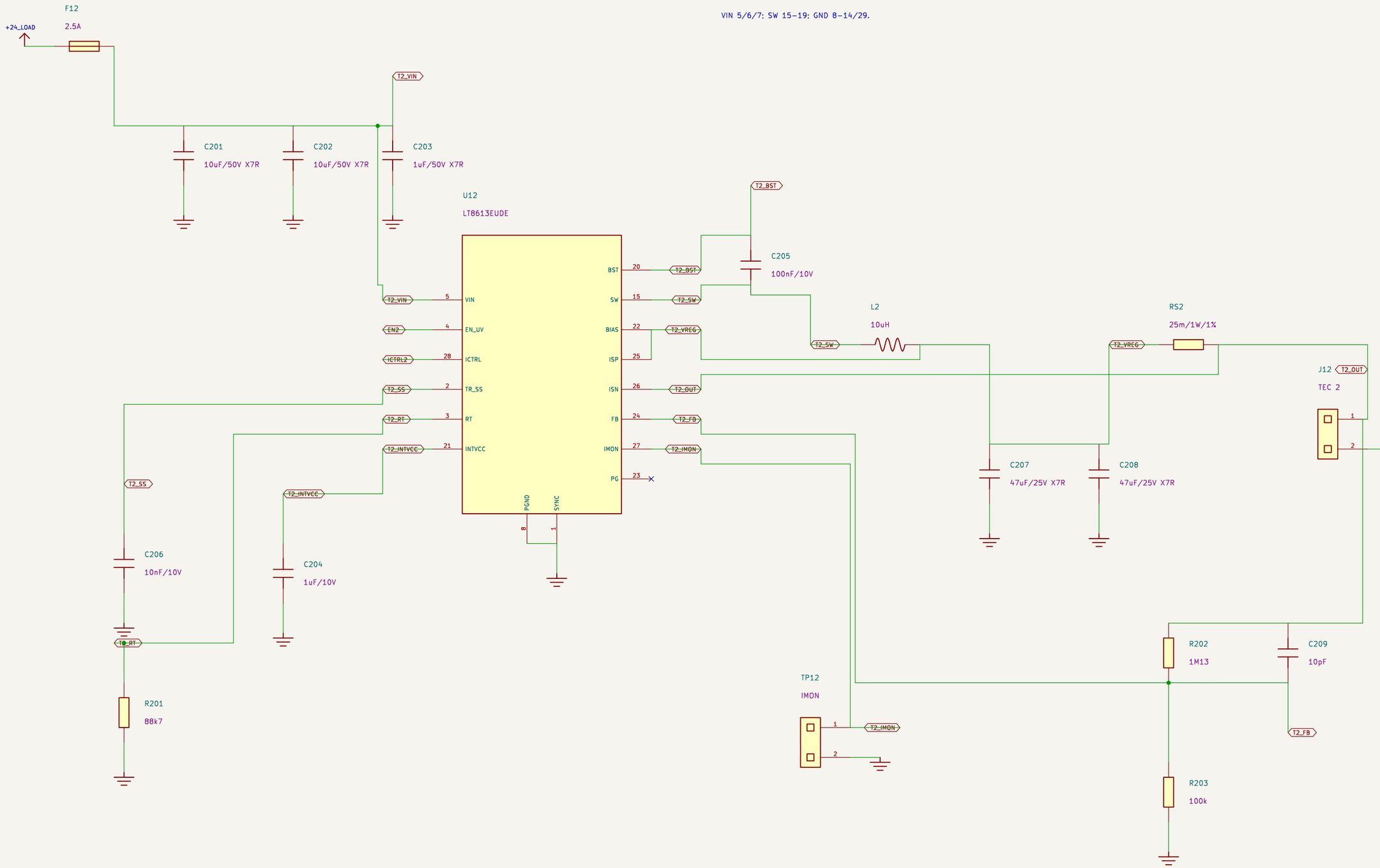


08 TEC current regulator 2

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB

Grouped IC pins:

VIN 5/6/7; SW 15–19; GND 8–14/29.



09 TEC current regulator 3

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB

Grouped IC pins:

VIN 5/6/7; SW 15–19; GND 8–14/29.

+24_LOAD

F13
2.5A

T3_VIN

C301
10uF/50V X7R

C302
10uF/50V X7R

C303
1uF/50V X7R

U13
LT8613EUDE

T3_BST

C305
100nF/10V

L3
10uH

RS3
25m/1W/1%

J13 T3_OUT
TEC 3

C307
47uF/25V X7R

C308
47uF/25V X7R

T3_SS

C306
10nF/10V

T3_INTVCC

C304
1uF/10V

T3_RT

R301
88k7

TP13
IMON

R302
1M13

C309
10pF

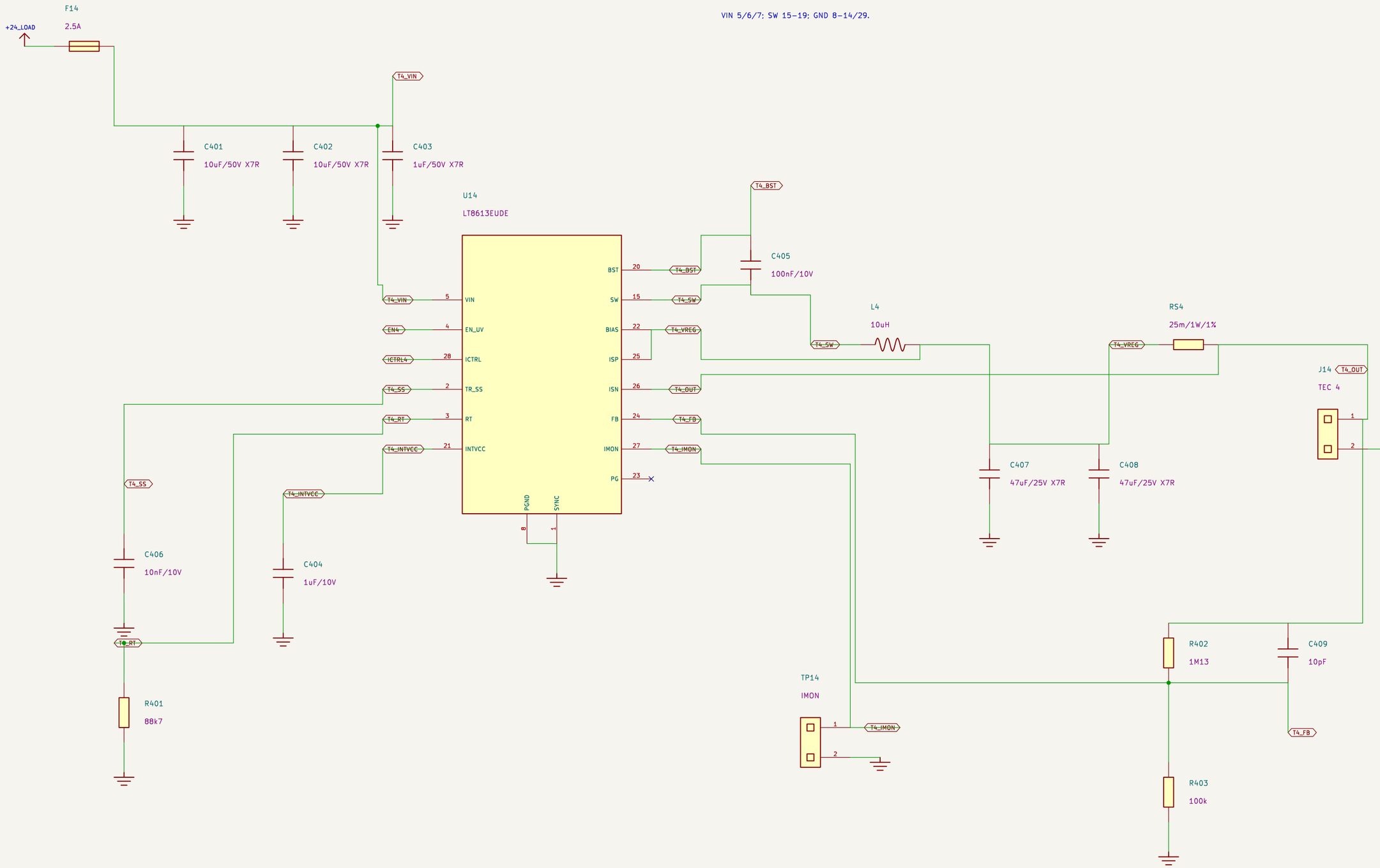
R303
100k

10 TEC current regulator 4

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB

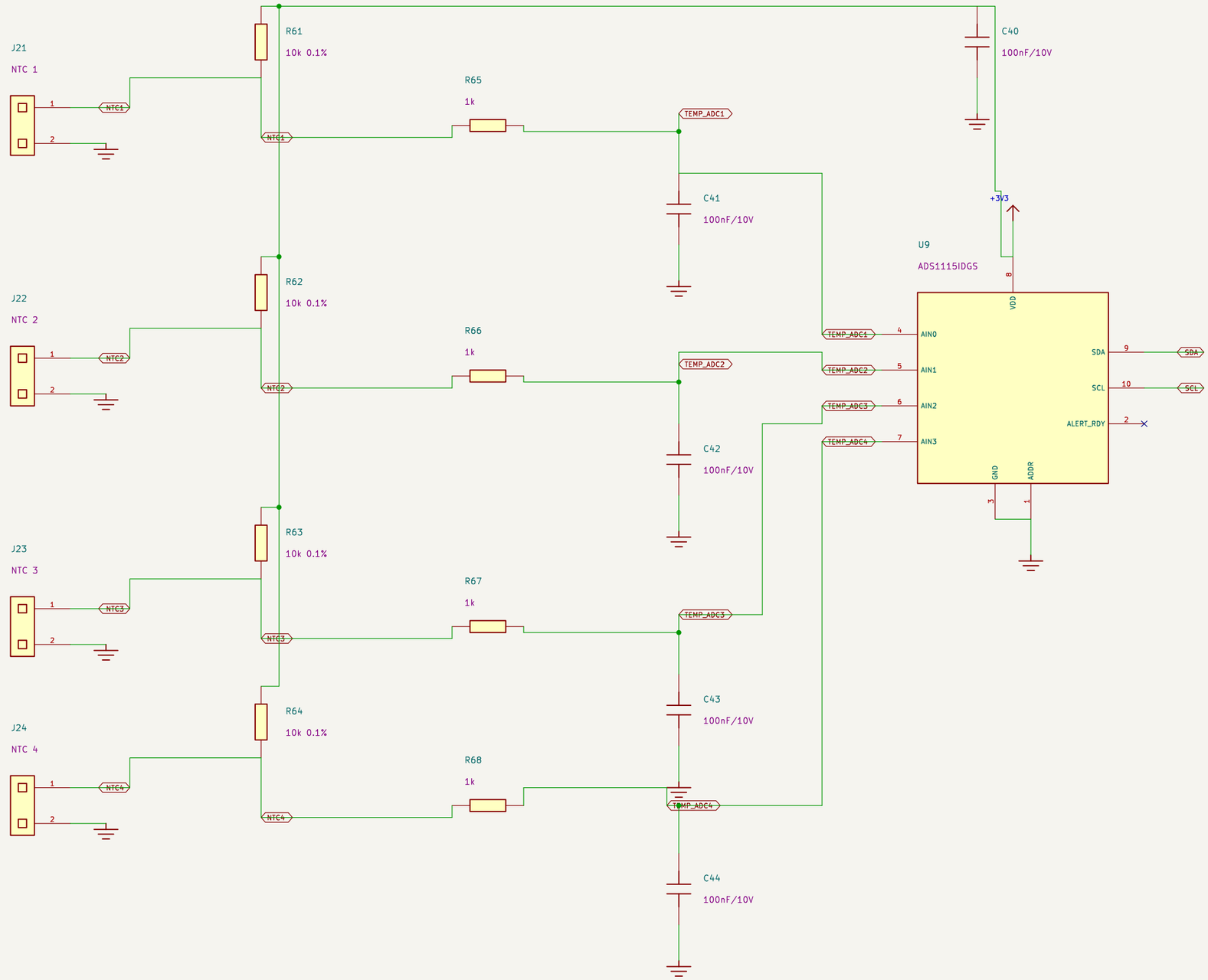
Grouped IC pins:

VIN 5/6/7; SW 15–19; GND 8–14/29.



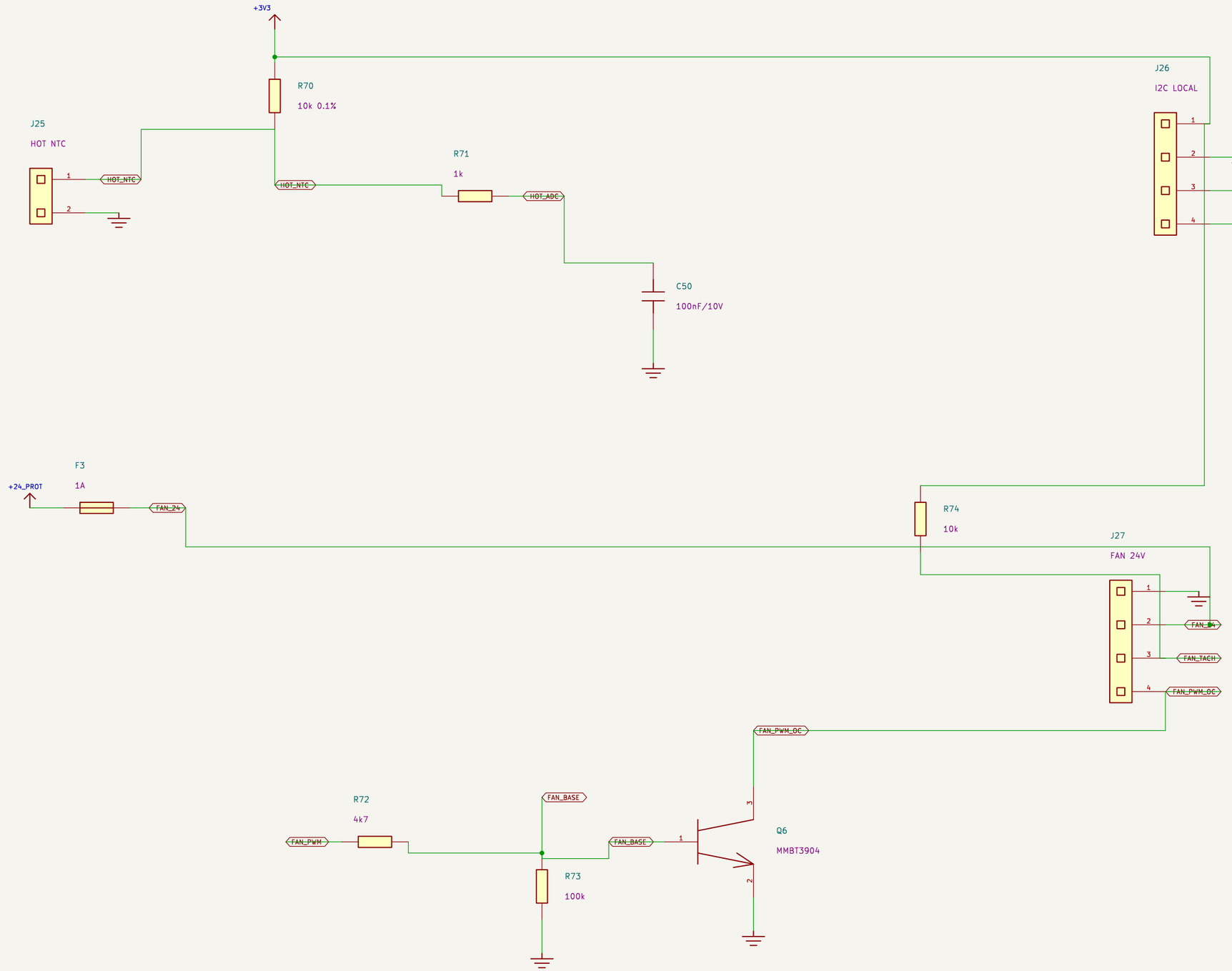
11 Glass temperature measurement

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB



12 Heatsink, humidity and fan

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB



13 Remote ambient sensor ports

ITO controller L | 24 V | prototype 4 A | wiring matches existing PCB

